

David Marco Sierra

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EDUCATION

Universitat Politecnica de Valencia

Dual Degree in Mathematics and Business Administration (2:1 Hons)

- Relevant Coursework: Statistics and Econometrics, Numerical Analysis, Partial Differential Equations.

Valencia, ES

09/23 – 05/28

Cambridge House British International School

High School Diploma (GPA 9.44/10)

- Mathematics (A), Physics (A), Business (A), Spanish Literature (A).

Valencia, ES

09/21 – 06/23

WORK EXPERIENCE

GPTadvisor – Wealth Management AI

Institutional Sales & Business Development

- Delivered €90k in revenue by designing and executing a global sales strategy, securing two POC deals.
- Contributed to sales meetings with C-suite executives at major financial institutions (e.g., BlackRock, Invesco).

Valencia, ES

06/25 – 09/25

Summer Intern

- Optimised AI model prompts, reducing response time by 5 seconds and enhancing user experience.
- Managed datasets of 70,000+ financial instruments (equities, bonds, indices) to design client-specific Focus Lists, enabling banks to refine product offerings and improve client outcomes.

05/24 – 09/24

PwC – Audit & Consulting

Insights Day (Received Summer 2025 Internship offer)

- Advised PwC professionals on AI applications to enhance internal efficiency during Insight Day.
- Shadowed a PwC Partner, gaining insight into consulting workflows and client engagement.

Valencia, ES

11/24 – 11/24

EXTRACURRICULAR

UPV Investment Club

Software Director

- Led the organisation of the first McKinsey & Company workshop at my university, attended by 100+ students, featuring a consulting case study and expert consulting tips.
- Secured a partnership with G-Research, and organised a quantitative finance event with 120+ attendees.

Valencia, ES

01/25 – Present

Morgan Stanley

FX Trading Hackathon

- Built and deployed an FX trading algorithm as part of a team, ranking 1st with 186% higher PnL than the runner-up.
- Detected a market crash via volatility analysis and hedged against a 35% drawdown.

Paris, FR

09/25 – 09/25

PROJECTS

Cross-Asset Momentum Strategy (Personal Research)

- Researched, implemented, and backtested (2020–2024) a systematic cross-asset momentum strategy with institutional risk validation across equities and commodities.
- Delivered 189.54% total return (23.72% CAGR, Sharpe 1.193, max drawdown -15.40%) and validated robustness with 10,000-path Monte Carlo/bootstrap analysis (CAGR 95% CI: 8.07%–40.74%; capital-loss probability: 0.1%–0.4%).

Options Risk Modelling & Volatility Analysis

- Built a Black-Scholes pricing and Monte Carlo simulation tool to analyse option risk; developed heatmaps to visualise Greeks' sensitivities and pricing behaviour.
- Conducted a comparative analysis of implied vs. realised volatility for S&P 500 and Bitcoin (2014–2024), leveraging historical data and statistical modelling to assess market behaviour and option trading insights.
- Performed a quantitative study of Delta in options pricing, analysing its role in hedging, exploring Taylor approximations, and simulating Delta vs. Delta+Gamma hedging strategies.

ADDITIONAL INFORMATION

Languages: Spanish (native), English (bilingual, C1), French (basic), German (basic)

Programming: Python (NumPy, Pandas, Matplotlib, SciPy), MATLAB, LaTeX, R

Tools: AI agents (Claude, Codex), MCP workflows, skill-based automation

Interests: Martial artist (MMA and Brazilian Jiu-Jitsu), travelling, skiing and football